Innovation Sprint Code Dev Submission Guide

1. Team Introduction

Name that Bird

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2. Problem Statement (preferably 150–250 words)

In the modern world people have less time to get outside and enjoy themselves and are often used to being entertained through quick and easy technologies such as streaming and social media. As changes in climate occur and urban sprawl increases in response to population increases and improved living standards our landscape is rapidly changing and we need to make sure we are securing our green spaces for future generations. In order to make sure we secure these spaces we need people from all walks of life to feel connected to the environment and our local wildlife.

The other problem is that as people spend less time outdoors, not only are they less aware of the condition of the environment and less connected, their chances to learn about are reduced. Particularly as more and more people move to large cities for work and housing opportunities.

Finally the last problem is that people often think of the great doors as inaccessible spaces and that can be a barrier to a range of people preventing them from learning about and experiencing the outdoors.

3. Solution & SDG Alignment (preferably 200–300 words)

I think that my app idea can fall under a range of the 17 Sustainable Development goals (SDG) of the UN.

The main goals I considered as I developed the app prototype were #13 Climate action, # 15 Life on land # 4 Quality Education and #10 Reduce-inequalities

4. AI Tools & Usage *(preferably 100–200 words)*

The application leverages several distinct AI flows, each designed for a specific task. The flows are categorized as either **contextual** (requiring reasoning, rules, and external data) or **instructional** (following a direct command).

- `contextual-identify-bird-flow`: This is the core reasoning flow. It acts as an expert ornithologist, using the Gemini 2.5 Flash model to analyze the audio recording, location, and time of year. It is "contextual" because it uses a Genkit tool to fetch current weather data, applies a hierarchy of evidence (e.g., prioritizing a habitat photo over GPS data), and generates structured output for the app.

- `contextual-generate-theme-flow`: This flow acts as a UI/UX designer. It takes three inspirational words and a mood (e.g., "High Contrast") and uses Gemini 2.5 Flash to generate a complete, accessible color palette. It is "contextual" because it must balance creative color generation with strict accessibility and contrast ratio rules defined in its prompt.

- `instructional-generate-bird-image-flow`: This is a simple,

direct-command flow. It uses the Imagen 4 model to generate a photorealistic image of the bird identified by the main contextual flow. Its sole instruction is to create an image based on the bird's common name.

- `tts-flow`: This utility flow converts text strings into audible speech using a specialized Gemini TTS model. It is used to power the application's audio narration for accessibility, providing spoken descriptions of pages and UI elements.

- **In-Progress AI Features**: Development is underway for an AI-powered assistant within the "Bird Book" log page. This assistant will allow users to filter, sort, and search their saved bird identifications using natural language commands in either text or voice.

5. Tech Stack Overview (preferably 75–150 words)

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## Tech Stack

- **Framework**: Next.js (App Router)
- **Language**: TypeScript
- **AI/ML**: Google's Gemini models (Gemini 2.5 Flash, Imagen 4) via Genkit
- **UI Components**: ShadCN UI
- **Styling**: Tailwind CSS
- **Database**: Firestore (for logging errors and future data storage)
- **Storage**: Firebase Storage (for audio recordings)
- **Deployment**: Firebase App Hosting

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6. Demo Description (preferably 150–250 words)

The demo currently has a few features set up and in the process of being connected to the gen AI and being debugged. The main focus is

the record button which uses gemini to identify birds based on their birdcalls.

7. Prototype and Github Links

Prototype link:

- Outdated now but this is what it looked like in first deployment if you want to see it is still active
- https://lyre-app-backend--studio-2836459659-ecbdf.us-east4.hosted.app/
-

Github link:

- Public until thursday
- https://github.com/MlKRW/studio.git
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8. Demo Video & Files (Video should not be more than 5 minutes)

- https://drive.google.com/drive/folders/1sdjG9Kr8juD9F-z7dXdMuD87zXcUQeDy?usp=sharing